Magnetic Holography

Optoelectronics (EEEN40640)

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This essay covers the paper “Magnetic Holography and Its Application to Data Storage” by Yuichi Nakamura. The paper provides a brief introduction to holography in general, followed by an overview of magnetic holography. A description of methods used to capture magnetic holograms is given, as well as several aspects which have been refined over the years in order to provide more accurate image recoveries (i.e., reduced signal-noise ratio) and to allow lower recording energies to be used. In this essay I will be giving an even briefer overview of these techniques based on my understanding of them and commenting on the overall accessibility and quality of the paper.

Holography is an information storage technique that records data using the interference and diffraction of light. This is done using a constant reference beam, and a signal beam encoding the data to be stored. The reference beam interferes with the signal beam, and the resulting interference pattern is recorded on some medium. This recording is called a hologram. The signal beam is recovered from the hologram by irradiating the hologram with the same reference beam used to create it. This happens via refraction of the reference beam through the interference pattern. This process allows high density storage data storage with fast recovery times, as large amounts of data can be stored in the same space and an entire 2D image of data can be recovered simultaneously.

The two main methods for forming and recovering holograms are the Two-Beam Interference method, and the Coaxial Holographic Interference method. In the two-beam method, the signal and reference beams are shone onto the medium at different angles, creating the interference pattern. In the coaxial method, the reference beam and signal beam are both passed through a Spatial Light Modulator, and the resulting beam is shone through a lens such that the medium is at the focal point of the lens. The former is most widely used, though the latter can be used with a similar optical system so a standard DVD or Blu-Ray Disc and has advantages against vibrations. Schematics of each method are given below, from the original paper.

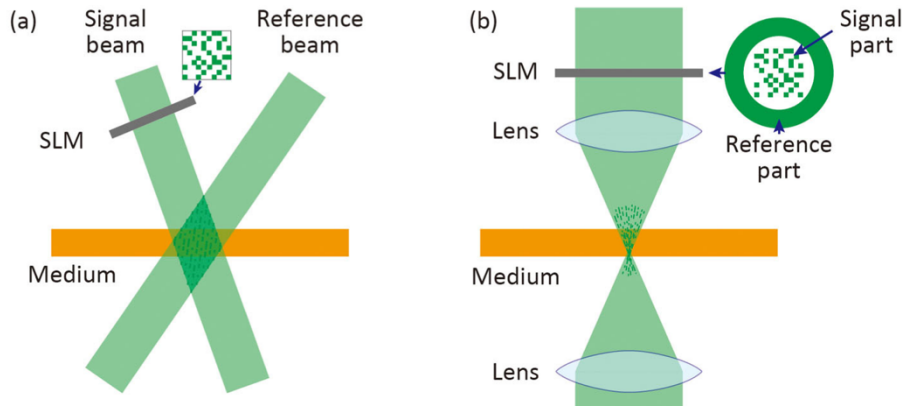

*Figure 1: Schematics of (**a**) two-beam interference method and (**b**) coaxial interference method.*

This paper covers in detail magnetic holograms, rather than traditional holograms. Magnetic holograms are stored as a difference in magnetization direction of the storage medium, as opposed to traditional holograms which are usually stored as a difference in refractive index of the storage medium. There are some issues regarding traditional holograms centred around potential volume changes in the material during the storage process while the photoreaction occurs. This can cause instabilities in the material. This

paper describes the use of magnetic garnets as a storage medium, which are very stable. With appropriately translucent garnets, volumetric holograms may be possible. The storage process is based on magnetic fringes being created by the interference pattern heating the garnets above their Curie Temperature (temperature at which a material loses its permanent magnetic properties) and allowing it to slowly cool and reverse the magnetization direction of the material. The signal is reconstructed by measuring the rotation of the plane of polarization of light incident on the hologram, caused by the Faraday Effect.

The holograms are made using the thermomagnetic recording method. This is the result of the light intensity distribution caused by the interference pattern heating up the recording medium. When the medium is heated above its Curie temperature its natural magnetization is lost. The external magnetic field is then used to reverse the direction of magnetization while the heated regions cool slowly. This region of reversed magnetization is the hologram. The heating process can cause problems if the medium is thermally conductive; if the heat dissipates away from the interference fringes too quickly then it can cause the fringes to become blurred, or even blend together into one if the recording energy is sufficiently high. This is especially a problem when metal films are used as the recording medium. If MnBi films are to be used as the recording medium, the paper reports that the pulse width of the laser should be on the order of 10 nanoseconds or less in order to avoid this issue. It is also reported that increasing the spacing between the fringes reduces the issue of the fringes bleeding together.

The polarization plane of linear polarized light falling on the medium rotates depending on the direction and the strength of the magnetization of the medium. This is the magneto-optical effect and is the principle behind the reconstruction of the signal beam. When the reference light is incident on the hologram, the light diffracted by the hologram interferes with the reference light, causing the reconstruction of the signal light. Nakamura gives the theoretical expression for the diffraction efficiency η of the first order diffracted light as:

$$\eta = \sin^2(\theta_F) \simeq \theta_F^2 = (Ft)^2 \tag{1}$$

Where θ_F is the rotation angle of the magnetic hologram, F is the Faraday rotation coefficient, and t is the depth of the magnetic hologram. This diffraction efficiency represents the brightness of the reconstructed image, which is improved by forming a deep hologram with a large rotation angle per unit thickness.

The paper reports on observations from the first demonstration of recording and reconstruction of magnetic holograms using coaxial interference. The first reconstructed image was dark and very noisy, so conditions were investigated to improve these conditions. One step was to remove the optical diffuser which was used to remove excessive concentrated recording energy. This contributed to the blurring of the recorded image. Modulation schemes for the 2D signal pattern were also investigated, and it was found that a pattern with the smallest white ratio was suitable for stable, error free recording with low recording energy.

The paper briefly touches multiplexing, which is one of the most important benefits of holography as a means of data storage. Multiplexing allows many different signals to be stored in the same physical space, which allows for a very high data storage density. This is done in traditional holography either using shift multiplexing (recording new signals at a slight physical offset), angular multiplexing (changing the incident angle of the light), and phase multiplexing. The only method that has been verified, as of this paper, is shift multiplexing. Four signal lights were recorded at $150\mu m$ shifts were recorded and successfully reconstructed. In the future it is hoped that this spacing will become much smaller, but this has demonstrated that the technique is feasible using magnetic holograms.

Since metals are often unsuitable storage media for magnetic holograms, transparent rare earth magnetic garnets are often used. This allows for greater hologram depth and better transparency. Despite having a low Faraday rotation coefficient, these materials allow higher rotation by using Magneto Photonic Microcavities. These consist of a transparent magnetic film sandwiched between two Bragg mirrors. This acts as an optical cavity and enhances the Faraday rotation angle. MPM recording media were demonstrated and were shown to greatly improve the diffraction efficiency of the holograms they created.

Another way to record a deep magnetic hologram is to increase the recording energy. However, this brings along the previously described issue in which the heat generated diffuses and causes the magnetic fringes to become blurred. Beyond a certain (quite low) point the increased recording energy stops providing any gains in efficiency and only serves to reduce the recoverability of the signal light. To combat this, media have been proposed which consist of alternating layers of magnetic material and heat dissipating material. These were demonstrated using numerical simulation and shown to be suitable, before being designed and fabricated for real testing. The Faraday rotation coefficient was slightly lower than initially expected due to nonidealities, but the diffraction efficiency was 1.6 times that of the single layer medium, which is thought to be the result of successfully creating deep magnetic fringes due to the ability to apply higher recording energy and better heat dissipation.

Overall, I found the paper to be extremely accessible, while also highly informative. The paper is well written using clear language and explains concepts in easy to parse terms. The diagrams and equations included help to demonstrate exactly the ideas behind the situations being described. It appears that holograms as a means of data storage is not exactly a novel idea, however, it is interesting to see if it can become more useful now that data storage is one of the most important challenges engineers face nowadays.

References:

[1]
Y. Nakamura, “Magnetic Holography and Its Application to Data Storage,” Photonics, vol. 8, no. 6, p. 187, May 2021 [Online]. Available: http://dx.doi.org/10.3390/photonics8060187